



ETEC510 DESIGN PROJECT

PHASE 2: CURRICULUM USER GUIDE

Critical Thinking Toolkit for High-Schoolers



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I. Introduction & Goals of this Guide

We present our Curriculum User Guide, which provides a high-level overview of developing a Gamified Critical Thinking Skills Course for educators, aligned with the BC curriculum and adaptable to other provincial curricula.

Our stated goals are to engage learners in a constructivist, relevant, differentiated, and design-based learning environment. The plan is to embed this in the curricular competencies of the BC Curriculum and the various aspects of the Core Competencies, fostering a community of learning and the development of a curriculum for 21st-century learners.

More specifically, we plan to provide a critical thinking toolkit for grade 10 students to understand and use. Our aim is for students to use these skills in real-life situations by providing a gamified knowledge progression course. At the end, students are expected to use these skills to develop cases of their own and challenge each other to develop true proficiency in these skills.

By creating this short course, teachers will be able to integrate the material into their curriculum planning and engage students in meaningful and purposeful ways. The tool can be integrated into either e-learning or in-person career studies courses, at the teacher's discretion.

II. Defining Critical Thinking

Shafer (2025) provides our tool with a strong theoretical framework that helps bridge the gap between BC competencies and our design. While Shafer does not offer a single explicit definition of critical thinking, he situates it within a blended perspective that integrates both generalist (skills-based) and specifist (contextual) understandings of critical thinking.

Therefore, in our view, critical thinking is defined as a set of interconnected skills that include reasoning, reflection, and socially responsible action, as seen through Shafer's framework. The pragmatic, dispositional, and transformational lenses provide a way to understand how these skills develop and interact.

Our tool adopts this same approach used in Shafer's (2025) research by combining both generalist and specifist perspectives. By combining these perspectives, we explore critical thinking not as a single skill, but as a set of skills that develop through practice, as evidenced through the various assessments and learning experiences embedded within our tool.

III. Basis of Pedagogy for the Course – *Participatory Culture*

Our toolkit is grounded in a participatory cultural pedagogy that supports students in developing critical thinking through situated practice. Situated practice is presented to students within a series of domains relevant to tenth-grade students such as relationships, academic decision-

making, and career life planning. Within these contexts, students are given the opportunity to analyze evidence, form judgments, and most importantly, use their critical thinking skills to mirror real-world participation.

Drawing on our collective understanding of Henry Jenkins et al.'s (2009) work on participatory culture, the toolkit is designed to give students opportunities to meaningfully exercise their agency within an engaging learning environment. Our goal is to provide “a safe space within which they can master the skills they need as citizens and consumers as they learn to parse messages from self-interested parties, and where they can separate fact from falsehood as they begin to experiment with new forms of creative expression and community participation” (Jenkins, 2009, p. 24).

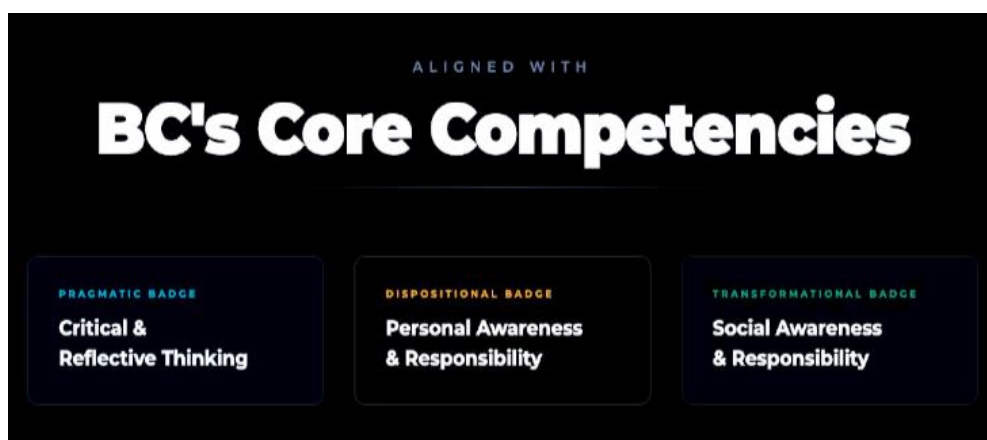
Our toolkit supports participatory culture by:

1. **Allowing students to exercise their agency:** Learners are given a space to intervene in scenarios, make decisions, and experience the outcomes of their choices.
2. **Valuing contribution:** Skill development takes time, and our design of this toolkit features badges that reward the progress the student makes. This offers a space for students to make mistakes in a low-stakes environment.

For educators, implementing this toolkit requires intentionally creating classroom conditions where participatory culture can emerge. This aligns with Jenkins et al.'s (2009) call for educators to provide spaces where young people can access the skills and experiences needed to become full participants.

In practice, supporting participatory culture means reducing barriers to engagement and expression. In accordance with Jenkins et al.'s (2009) understanding of participatory culture, this toolkit positions educators as guides who support students as they navigate scenarios grounded in their own lived contexts.

IV. Creating Course Content using Curriculum Competencies – *BC curriculum connection*



This section maps the toolkit to the BC Core Competencies framework, demonstrating how it supports the transferable capacities BC expects students to develop across all courses and grade levels. While Career Life Education 10 is our proposed primary delivery context, the Core Competencies framing makes the case for the toolkit's broader curricular value across subject areas, grade levels, and provinces. Cross-provincial equivalencies are available through the CERIC Career Curriculum guide (careerwise.ceric.ca).

The BC Core Competencies: An Overview

BC's Core Competencies are transferable capacities that students build across every subject and grade level, from kindergarten through graduation. They are not a separate course — they are woven into all learning, documented through teacher observation and student self-reflection, and reported on as part of the standard assessment cycle.

BC identifies three Core Competency areas:

- **Communication** — connecting with others, acquiring and presenting information, collaborating
- **Thinking** — subdivided into Creative Thinking and Critical & Reflective Thinking
- **Personal & Social** — subdivided into Personal Awareness & Responsibility and Social Awareness & Responsibility

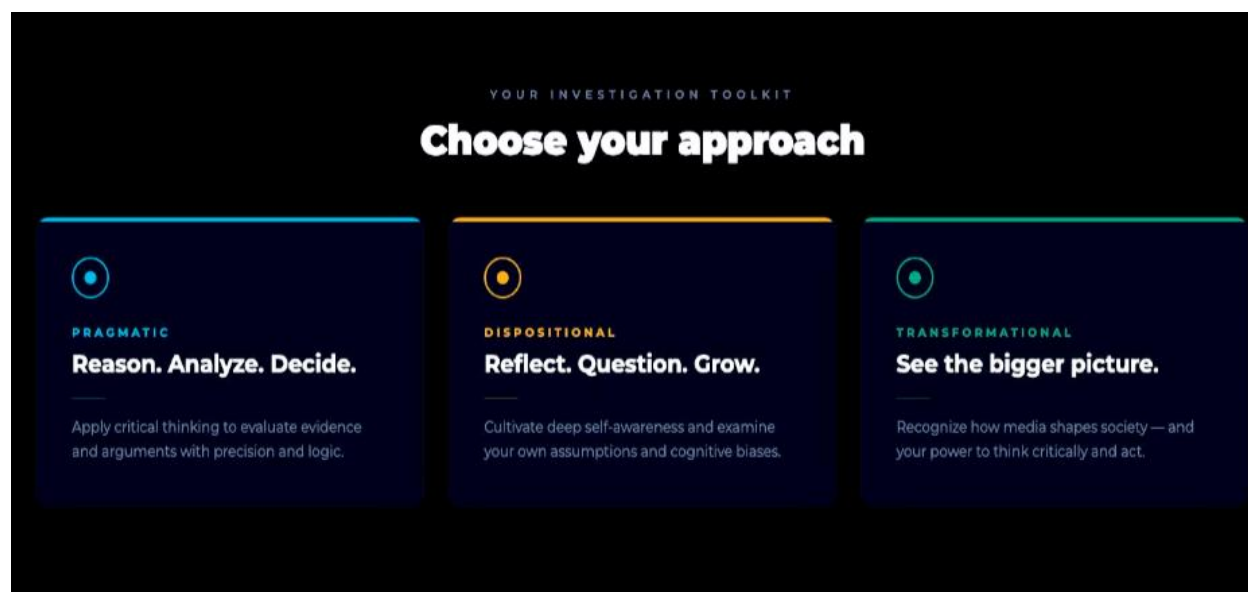
Because Core Competencies cut across all subjects, any tool that develops them has curricular legitimacy in multiple classrooms — not just one course. A critical thinking toolkit that explicitly targets these competencies can be used in English Language Arts, Social Studies, Career Education, or any other subject where teachers are looking for structured ways to document and develop student thinking.

Key point: *Core Competencies are assessed through observable student behavior, not tests. This means a gamified, activity-based toolkit is a particularly well-suited vehicle for developing and documenting them*

Mapping the Three Lenses to BC Core Competencies

The toolkit's critical thinking framework is based on Shafer's three-lens approach. The direction for teachers is to produce course content based on this scholarly approach.

Shafer's (2025) three-lens critical thinking framework — pragmatic, dispositional, and transformational — maps directly onto BC's core competency clusters. Each badge students earn in the toolkit corresponds to a lens and to a distinct competency area. The third column in the chat below outlines possible strategies teachers can leverage in their own unit planning to align the toolkit to further develop a BC core competency.



Badge / Lens	BC Core Competency	Possible Strategies
Pragmatic (Reasoning) 	Critical & Reflective Thinking — questioning assumptions, analyzing evidence, forming and defending judgments	• Show students two versions of the same information and ask them to decide which to trust • Help students see how outside forces — ads, algorithms, trends — shape what information they see • Ask students to follow a claim back to where it started

Dispositional (Reflection) 	Personal Awareness & Responsibility — self-regulating, reflecting on learning, developing identity and values	• Ask students to explain how they reached their decision, not just what they decided • Show students how an expert approached the same problem, without framing it as correction • Let students track their own progress so they can notice how their thinking is changing
Transformational (Agency) 	Social Awareness & Responsibility — contributing to community, considering consequences for others, valuing diverse perspectives	• Put students in situations where their choices affect others, not just themselves • Have students work together to reach a decision they each have to defend • Ask students to create something — a challenge, an argument, a case — that a peer will engage with
All lenses	Communication — explaining and justifying thinking, collaborating, acquiring and presenting information	• Let students speak their thinking out loud before they have to write it down • Have peers identify one specific strength in each other's reasoning, not just whether they agree • Use multiplayer tasks that require students to negotiate and communicate across different perspectives

Possible Area to Explore: Aligning with BC Proficiency Scale

BC describes Core Competency development using four proficiency levels: Emerging, Developing, Proficient, and Extending. One direction worth exploring is whether the toolkit badge and progression system could be mapped to this scale with students moving from structured, scaffolded tasks toward independent application, roughly mirroring the shift from Emerging to Extending. The table below sketches out what this might look like and is offered as a starting point for further discussion rather than a finalized design decision.

Proficiency Level	What It Looks Like in BC's Framework	Where the Toolkit Addresses This
Emerging	Student is beginning to identify and apply the competency with significant support	Early guided missions with visible frameworks, hints, and expert modeling

Developing	Student applies the competency with some support and is growing in consistency	Mid-level missions with scaffolding that fades as students demonstrate confidence
Proficient	Student applies the competency independently and consistently in familiar contexts	Badge completion tasks requiring independent judgment across a full case
Extending	Student applies the competency in new contexts and supports the learning of others	Sandbox phase and peer creation tasks where students design challenges for classmates

Cross-Curricular Delivery Options

Because Core Competencies belong to every subject, this toolkit has a legitimate home in multiple classrooms. The table below identifies subject areas where teachers could integrate it and the primary competency each context would emphasize.

Subject Area	Primary Competency Emphasis	Natural Entry Point
Career Life Education 10	Personal & Social; Communication	Self-Assessment or Workplace Readiness units; digital footprint content
English Language Arts 10	Critical & Reflective Thinking; Communication	Media literacy, persuasion, and source analysis units
Social Studies 10	Critical & Reflective Thinking; Social Awareness & Responsibility	Civic participation, media and democracy, or current events units
Applied Design, Skills & Technologies	Creative Thinking; Communication	Design thinking, iterative problem-solving, or digital literacy modules
Any subject (advisory/homeroom)	All Core Competencies	Stand-alone module delivered during advisory, flex, or study blocks

A Note on Pedagogy: Cognitive Apprenticeship as One Possible Framework

Teachers implementing this toolkit may find cognitive apprenticeship (Collins, Brown & Newman, 1989) a useful lens. The framework proposes making expert thinking visible to learners through modeling, coaching, scaffolding, and gradually releasing responsibility — a sequence that maps naturally onto the toolkit's own progression from guided missions to open

sandbox. This is one possible implementation approach, not a requirement; teachers should adapt the tool to their own pedagogical context and subject area.

Reference: Collins, A., Brown, J. S., & Newman, S. E. (1989). *Cognitive apprenticeship: Teaching the crafts of reading, writing, and mathematics*. In L. B. Resnick (Ed.), *Knowing, learning, and instruction* (pp. 453–494). Lawrence Erlbaum.

Cross-Provincial Applicability

The Core Competencies framework is BC-specific, but the competencies themselves - critical thinking, communication, personal and social responsibility - are shared goals across Canadian provincial curricula. The three-badge structure (reasoning, reflection, agency) maps broadly onto career development, personal planning, and media literacy courses nationwide. Teachers outside BC can use the CERIC guide (careerwise.ceric.ca) to identify the closest local equivalents.

V. Technology Requirements for Developing & Integrating the Course

Optimized for Personal Computer (PC)

The intended use case for this curriculum toolkit is best optimized for a PC. Given this is a course based approach, students should be expected to engage with the content on a larger screen, as there is both a **reading & comprehension** component, along with **engagement**.

The required technology for this includes:

- Personal Computer or Laptop
- Stable internet connection (recommended minimum of 10mb/s)
- Mouse, keyboard

Step 1) Reading & Comprehension – Learning Management System (LMS)

This toolkit is designed to allow flexibility for teachers to use their LMS of choice. We suggest Moodle.

Moodle is a Learning Management System (LMS) that will be used to house the course in its entirety. Educators will be able to develop the course through this LMS, and tailor the content for their students.

<https://moodle.org/>

Moodle is able to handle:

- Course Setup
- Content Library
- Enrolment
- Grading
- Progress
- Access & Security

A full user guide, from start to finish, in setting up content, course, student & grading can be found at the “Moodle Academy” from the company itself:

https://www.youtube.com/watch?v=SrNXFPPhsPk&list=PLxcO_MFWBDfJRhsDTNPA5IwesdCMKF4d&index=3

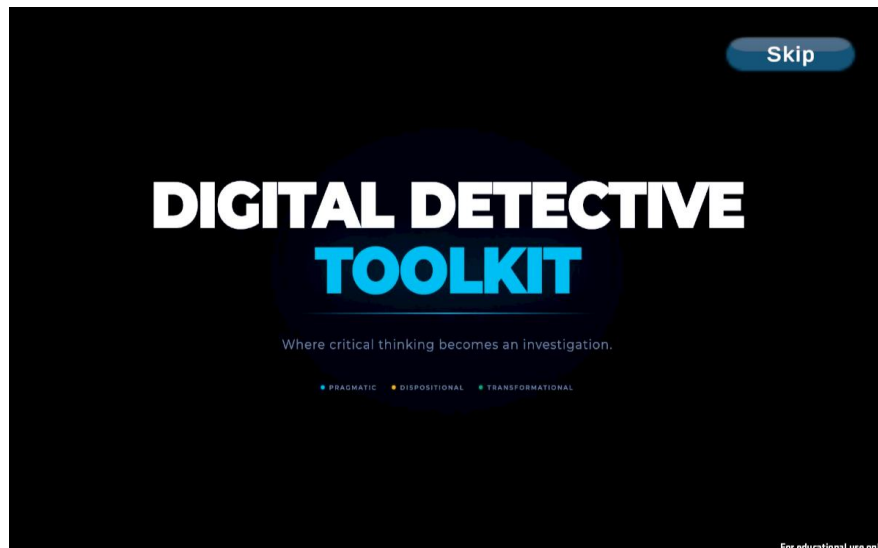


Keys for success in using Moodle:

- **Use Moodle as a library first:** Use your LMS as a place to hold readings, videos, tutorials and all content you want your students to engage with

- **Provide a clear starting point:** Include a "Start Here" section or an introductory block at the top of the course with the syllabus, course schedule, and an introductory forum.
- **Use the "Hidden" feature during development:** Keep sections or activities hidden from students while you are still drafting or editing them to avoid confusion.
- **Provide clear instructions:** Always include explicit directions, due dates, and expectations within the description box of every assignment or activity.
- **Enable Activity Completion:** Turn on completion tracking so students can see checkmarks next to completed tasks, helping them visually track their progress through the course.
- **Use Restrict Access strategically:** Set conditions that require students to achieve a certain score on a quiz or view a specific resource before the next module unlocks.

Step 2) Engagement – Gamified Toolkit – Unity



Once the course has been developed, including the learning content, a link can be provided to students. This will allow them to access the gamified toolkit to engage with the learnings in the intended environment.

Unity is the gamification engine behind the course's learning objectives.

<https://genially.com/>

Genially will be used for:

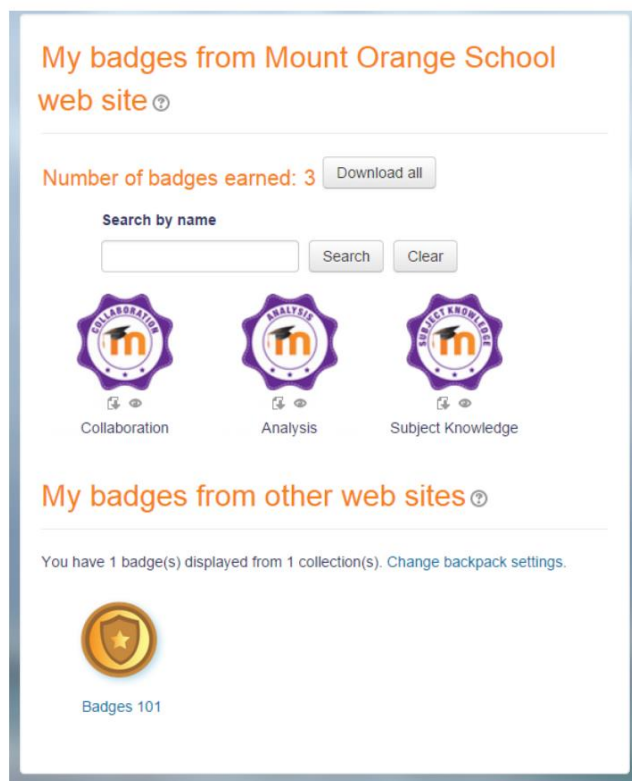
- Interactive Content
- Animations and Scenario Building
- Infographics & Presentation

<https://genially.com/features/>

Connecting Moodle & Unity Progression: Badges

Most LMS will have a badge system as a positive reinforcement item. Given that the Unity toolkit is badge reliant as a progression system, it is necessary to connect the badge collection in your LMS with the toolkit's progression.

Using Moodle as an example, here is what the student will see as their progression:



Technology Accessibility

For scenarios where students require accessibility needs, the following can be modified for your use case.

Teachers can leverage the robust, built-in features of their school's existing Learning Management System (LMS), whether that is Moodle, Canvas, D2L Brightspace, or Google Classroom, to act as an accessible "wrapper" around the Digital Detective toolkit.

By embedding our gamified module within a standard LMS, teachers can seamlessly integrate Universal Design for Learning (UDL) principles and further reduce barriers. The LMS can expand options for means of engagement, representation, and expression when there are limited options on the gamified tool. For example, while information is relayed through videos and visual clues in the game, teachers can also provide written transcripts or summaries within the LMS for learners who require additional processing time. Along with that, learners can express their learning in many ways which are already embedded in the game, such as recording voice memos and writing responses, but more opportunities can be offered through the LMS if needed. This is crucial for fostering a truly participatory culture in which all students feel capable of contributing.

Offline Usage:

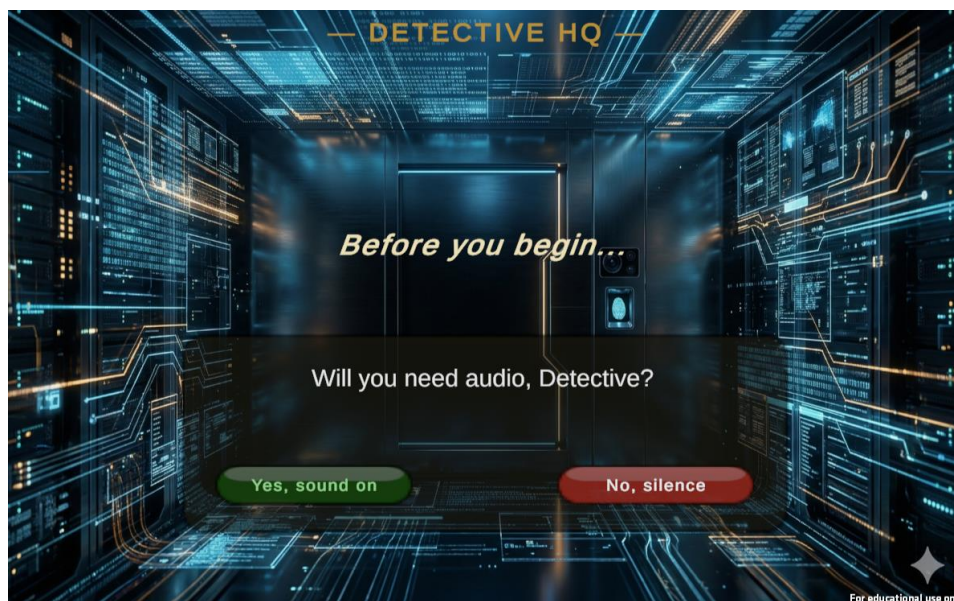
Content from Moodle can be issued as PDF documents and further printed if needed.

The unity game can be downloaded, and although all features, such as the collaborative elements and teacher tracking, cannot be used offline, it may provide students an opportunity to learn through the experience.

**It would be recommended to have the student write or provide a verbal follow -up if offline use is required, as a gap between the teacher tracking component.*

Hearing options:

The Unity toolkit has transcription built-in, where students can choose to have audio or not.



The LMS can house downloadable documents or rich-text transcripts of all in-game video briefings, audio clues, and case files. This ensures that students with hearing impairments, language-processing needs, or those who simply prefer reading to listening have equal access to the evidence.

Visual Support:

Screen Reader Compatibility: Most modern LMS platforms natively support web accessibility standards (WCAG) and screen readers. Teachers can use the LMS page surrounding the game to provide screen-reader-friendly alt text or text summaries of key visual cues, ensuring visually impaired students are not excluded from the investigative process.

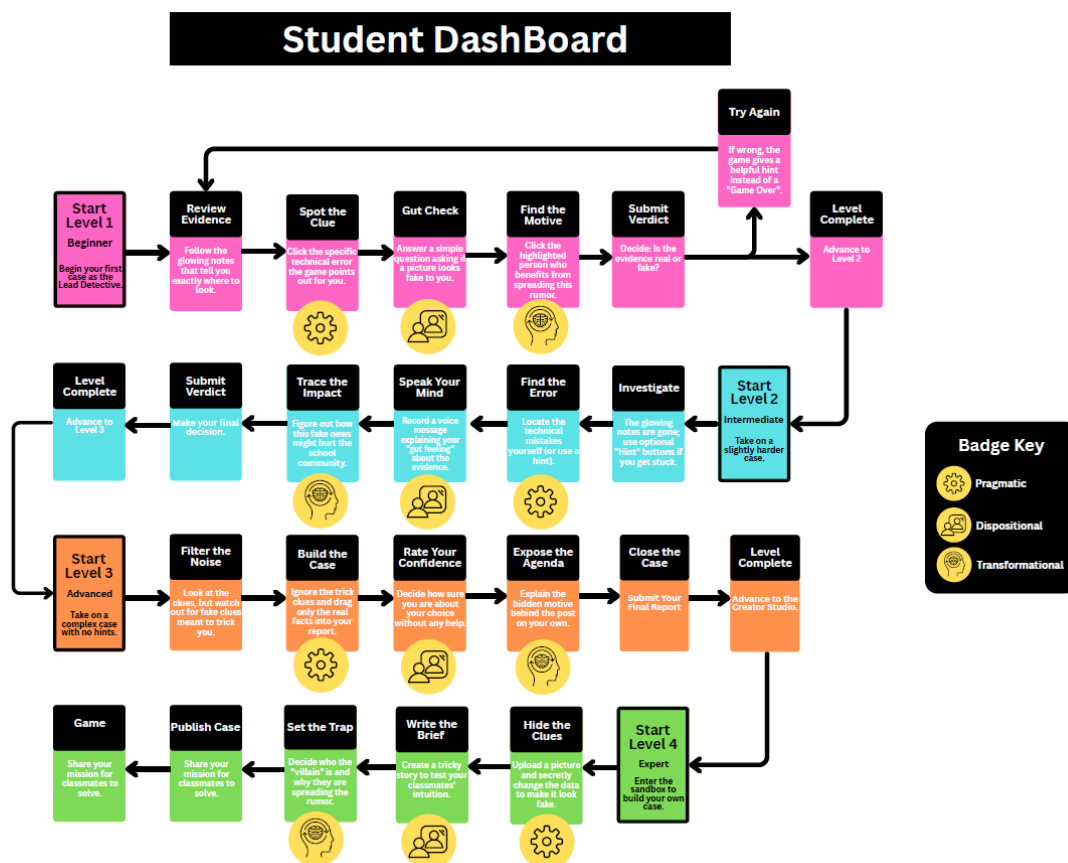
Overload Concerns:

Alternative Submission Pathways: While our in-game Voice Memo feature is designed to reduce writing barriers, we recognize that some students may face speech-related challenges or anxiety. The LMS provides built-in text submission boxes or private messaging systems, allowing students to submit typed reflections or video logs to earn their Dispositional and Transformational badges outside of the primary game interface.

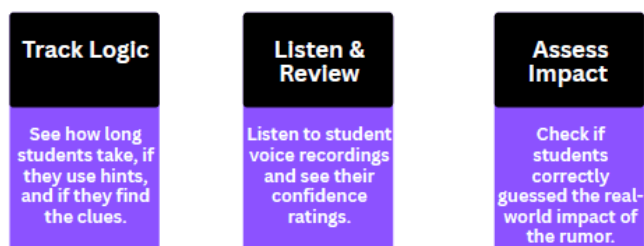
Adaptive Pacing and Cognitive Chunking: Grade 10 learners often vary in their executive functioning and ability to manage cognitive load. Educators can use the LMS's module release conditions or built-in checklists to "chunk" the game into smaller, manageable missions. This allows students to tackle one badge or phase at a time without feeling overwhelmed by the entire case file. All video scenes also include a replay option, allowing students to review information at their own pace.

Replay Message

V. Visual Progression User Guide for the Unity Engagement Toolkit



Teacher Dashboard



Visual Progression User Guide: The Learner's Journey

[Click this link to access the toolkit as you walkthrough the visual progress user guide.](#)

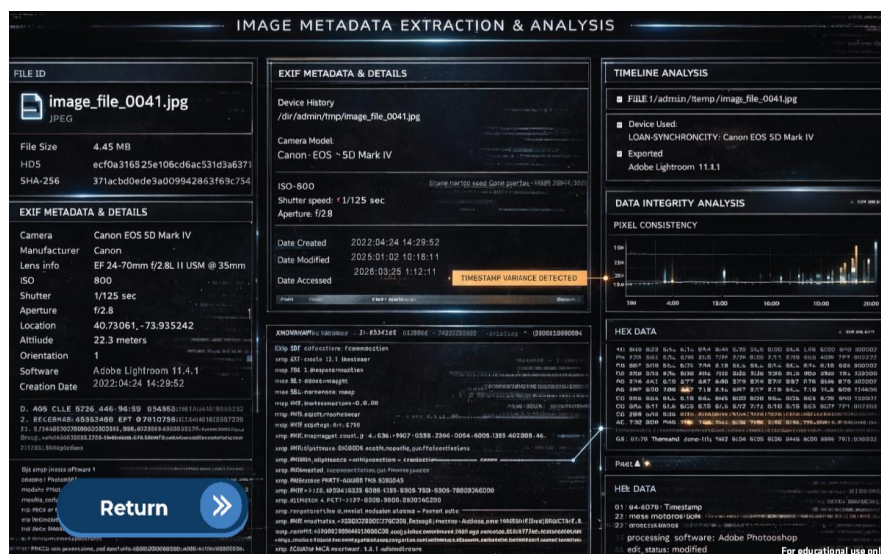
The Digital Detective toolkit is designed as a dynamic, interactive learning environment where progression is based strictly on cognitive complexity. In order to advance, each mission requires the student to successfully fulfill and earn three specific critical thinking badges Pragmatic, Dispositional, and Transformational at their current level of complexity.

Phase 1: Beginner

In the initial stage, students log into the Moodle Learning Management System and launch the interactive Genially module. At this beginner level, the interface provides significant expert modeling, explicitly guiding students toward earning their first set of badges.

Mechanics & Badge Acquisition: Students are introduced to the Evidence Board, where glowing "Detective's Notes" explicitly direct their attention.

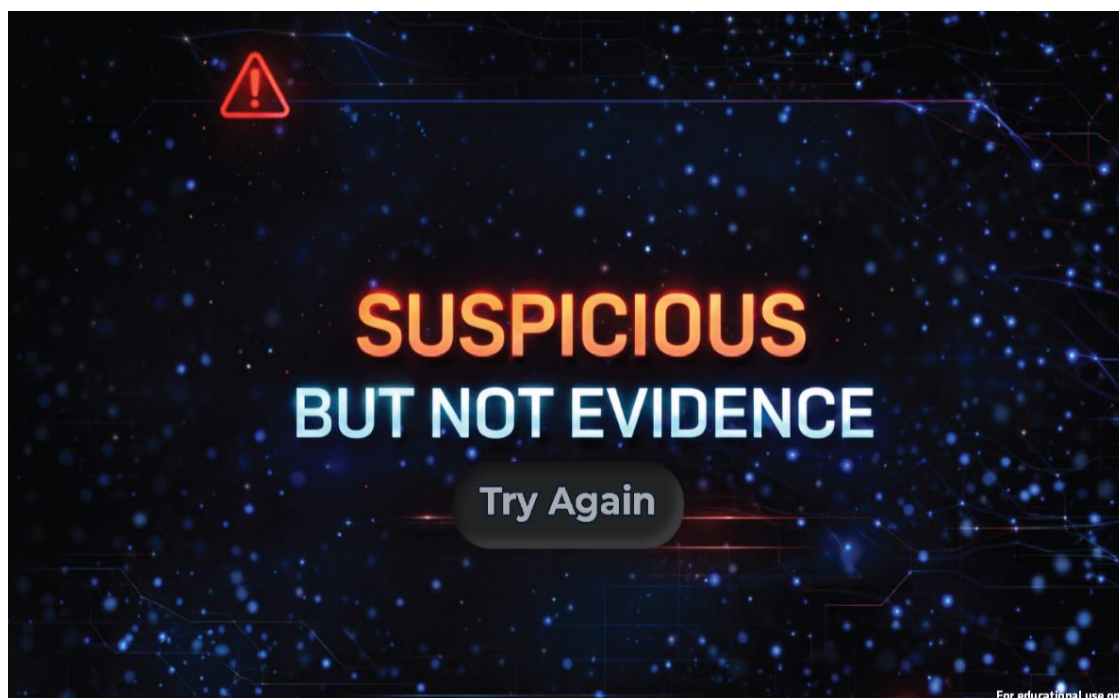
- **Earning the Pragmatic Badge:** Students must click explicitly highlighted technical clues, such as a flagged software tag in the metadata log.



- **Earning the Dispositional Badge:** Students are prompted with a highly scaffolded "gut check" question, asking them to select whether a visual anomaly (like six fingers on a hand) feels unnatural.
- **Earning the Transformational Badge:** The UI will specifically prompt a student to use the Transformational lens to ask who benefits from a piece of media being shared.

Glowing notes guide students to identify the specific beneficiary of a fake post, such as a brand-new social media account seeking engagement.

Participatory Culture Integration (Safe Failure): To foster a safe space for skill mastery without the fear of penalization, this level features a built-in "Redemption Mechanic." If a student makes an incorrect judgment or submits the wrong final verdict, they are not met with a "Game Over." Instead, they are gently redirected back to the evidence to try again, reinforcing that mistakes are a helpful and necessary part of the learning process.



Phase 2: Intermediate

As students enter the intermediate phase, the direct instructional scaffolding begins to fade, requiring more active cognitive engagement.

Mechanics & Badge Acquisition: The explicit "Detective's Notes" are removed. In their place, the UI offers optional "Hint" buttons. Students must now begin to rely on their own internal reasoning to apply Shafer's three thematic lenses.

- **Earning the Pragmatic Badge:** Students must locate visual and technical anomalies using the optional, user-triggered "Hint" buttons if they get stuck.
- **Earning the Dispositional Badge:** To support diverse communication styles and reduce barriers to expression, students must use a "Record Voice Memo" feature before submitting their final verdict. They earn this badge by speaking their reasoning aloud, allowing them to articulate their thoughts authentically without being hindered by spelling-related self-consciousness.

- **Earning the Transformational Badge:** Students must analyze a scenario and identify the broader social consequences of the media spreading, moving beyond just identifying the original poster.

Phase 3: Advanced

In the advanced phase, the training wheels are entirely removed. Students are expected to apply the BC Core Competencies independently in the presence of distracting information.

Mechanics & Badge Acquisition: The Evidence Board introduces "Red Herrings" pieces of evidence that are irrelevant or purposefully misleading.

- **Earning the Pragmatic Badge:** Students must independently filter out irrelevant clues, successfully drag the relevant evidence into their "Case Closed" report, and justify their reliance on the correct technical facts.
- **Earning the Dispositional Badge:** Students rate their own confidence level on a sliding scale and defend their stance without any prompting or hints.
- **Earning the Transformational Badge:** Students independently predict the community impact and hidden agendas behind the media campaign.

Critical Application: Earning badges at this level signifies a strong, independent grasp of digital media literacy, as students have successfully and independently articulated their Pragmatic, Dispositional, and Transformational justifications.

Phase 4: Expert

The final phase of the flowchart represents the pinnacle of Jenkins' (2009) Participatory Culture. The student transitions from a consumer of the game to a creator.

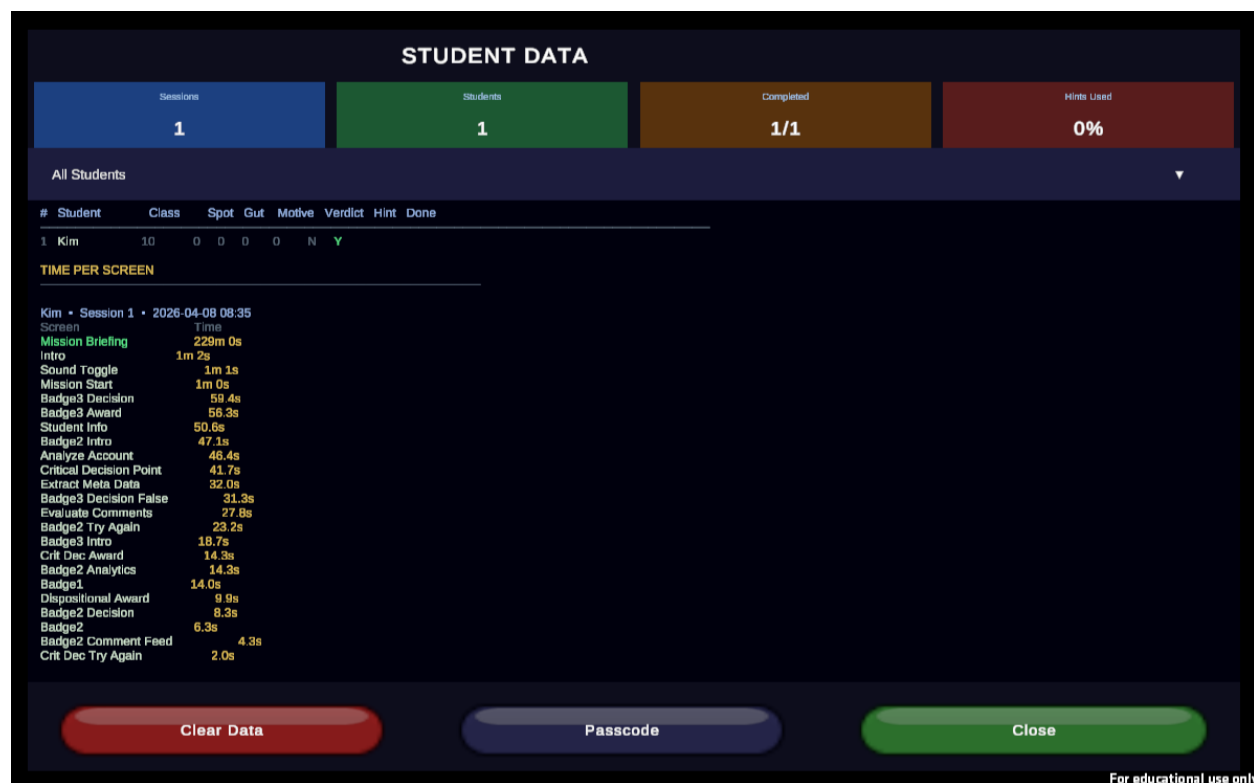
Mechanics & Badge Application: The UI shifts to a "Case Creator" level editor. Students use a sandbox approach to upload their own media, plant visual anomalies, manipulate metadata, and assign badge triggers.

- **Applying the Pragmatic Skill:** Students intentionally plant logical anomalies or alter metadata in the media they upload.
- **Applying the Dispositional Skill:** Students write the case briefing to intentionally test the intuition and biases of their peers.
- **Applying the Transformational Skill:** Students design the overarching social scenario and hidden motives that their classmates will need to uncover.

Social Learning: By publishing these custom cases for their peers to solve, students not only demonstrate extended mastery of the subject matter but also actively support the learning of their classmates, fulfilling the highest tier of the BC Proficiency Scale. This grants them the ultimate "Lead Detective" status.

The Backend: Teacher Dashboard

Running parallel to the learner's journey is the Teacher Assessment Dashboard. As the student interacts with the Genially modules, data is securely pushed to the educator. The dashboard visualizes the student's badge acquisition, tracks time spent on evidence, logs hint usage, and stores their recorded voice memos. This provides the educator with observable, concrete data points to apply grading rubrics accurately, ensuring the gamified experience remains deeply tied to measurable curricular outcomes.



V. Assessing Student Engagement and Performance

Formative and Summative Tasks

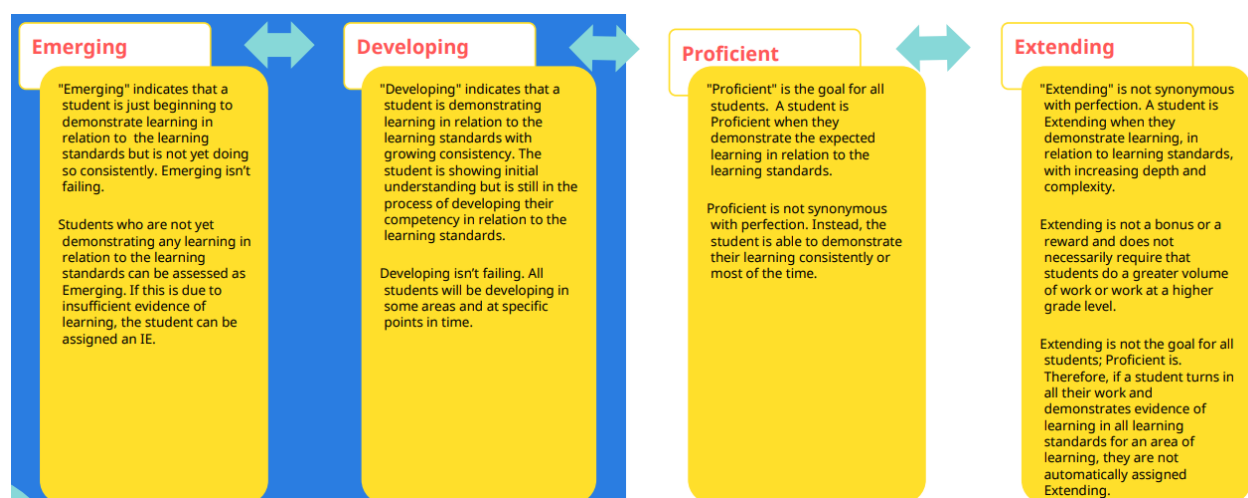
Students will be assessed through formative and summative tasks. The priority of formative assessments will be to check in on learning to guide students on their journey through the material. The tasks can include informal observations and low-stakes quizzes. Summative assessments are meant to assess final learning to examine what students have taken from the module and what skills they can demonstrate. Below is a list of suggested assessments.

Formative Assessment	Summative Assessment
- Activities between earning badges for practice - Progress tracking to see whether or not students are on the right path	- Ability to navigate through summative tasks to gain badges through explaining reasoning - Predicting hidden agendas

- | | |
|--|--|
| <ul style="list-style-type: none"> • Observations of student conversations and collaborations to come to new conclusions and perspectives • Informal check-ins – short quizzes for understanding | <ul style="list-style-type: none"> • Voice Memo Justifications • Analysis of scenarios • Participation in Case Studies and activities |
|--|--|

Grading

Assessment is based on the BC Proficiency Scale rather than quantitative scoring. Instead of accumulating points, educators will use rubrics to evaluate qualitative observations of a student's skill acquisition. For instance, a student's grade is determined by their ability to demonstrate reasoned thinking in the gamified summative tasks. When using this tool, educators are encouraged to grade based on the learning tasks in levels 3 and 4. This ensures that students have been given the opportunity to practice in earlier levels to support authentic assessment.



Unpacking the proficiency scale support for educators. Government of British Columbia. (n.d.). <https://www2.gov.bc.ca/assets/gov/education/administration/kindergarten-to-grade-12/unpacking-the-proficiency-scale-support-for-educators.pdf>

Sample Rubrics

Students will be graded based on the guidance of a rubric to assess their proficiency. Educators using this tool can create their own rubrics, but below are some samples of the expected outcomes of curricular competencies across a variety of subject areas.

Career Education 10

	Emerging	Developing	Proficient	Extending
Identify risks and appreciate	Identifies basic consequences	Identifies consequences	Identifies consequences	Identifies consequences

benefits associated with personal and public digital footprints	of digital footprint with significant support using the transformational lens.	of digital footprint with some support using the transformational lens.	of their digital footprints on themselves and others using the transformational lens	of their digital footprints in complex scenarios using the transformational lens.
Identify career-life challenges and opportunities, and generate and apply strategies	Identifies basic motives of digital media with significant support using the pragmatic lens.	Identifies underlying motives of digital media with some support using the pragmatic lens.	Identifies underlying motives of digital media using the pragmatic lens.	Identifies a variety of underlying motives of digital media using the pragmatic lens.

New Media (English Language Arts) 10

	Emerging	Developing	Proficient	Extending
Think critically, creatively, and reflectively to explore ideas within, between, and beyond texts	Reflects on their own thinking process with significant support using the dispositional lens.	Reflects on their own thinking processes with some support using the dispositional lens.	Reflects on their own thinking processes in familiar contexts using the dispositional lens.	Reflects on their own thinking processes in complex contexts using the dispositional lens.
Respectfully exchange ideas and viewpoints from diverse perspectives to build shared understanding and extend thinking	Explains their positioning with minimal detail begins to listen to opposing perspectives using the transformational lens.	Explains their positioning on familiar contexts with support and listens to opposing perspectives using the transformational lens.	Explains their positioning in familiar contexts with clarity and understands opposing perspectives using the transformational lens.	Explains their positioning in complex contexts with clarity and understands and can connect to opposing perspectives using the transformational lens.

Social Studies 10

	Emerging	Developing	Proficient	Extending
Make reasoned ethical judgments about actions in the past and present, and assess appropriate ways to	Begins to apply reasoning on how outside forces can influence the presentation of information using the pragmatic lens.	Applies reasoning on how outside forces can influence the presentation of information with support using the pragmatic lens.	Applies reasoning on how outside forces can influence the presentation of information using the pragmatic lens.	Applies sophisticated reasoning on how outside forces can influence the presentation of complex information

remember and respond				using the pragmatic lens.
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VI. Summary of Guide

Youth are more engaged with digital media environments and must also develop important skills to participate critically and ethically. True to the participatory learning philosophy, this guide reinforces Jenkins' view: "a focus on expanding access to new technologies carries us only so far if we do not also foster the skills and cultural knowledge necessary to deploy those tools toward our own ends." (Jenkins, 2009, p. 8)

This idea directly informs the goals of this course regarding ways to develop and reinforce students' critical thinking skills to carefully engage with digital and media consumption and participation.

The course will provide students with opportunities for lifelong learning across various domains, including relationships, career pathways, and educational planning. Through its development, the course will aim to incorporate gamification elements that foster a reflective and collaborative learning environment while effectively integrating digital technologies.

References

BC Ministry of Education. (n.d.). Core competencies. BC Curriculum.

<https://curriculum.gov.bc.ca/competencies>

BC Ministry of Education. (n.d.). Career-life education. BC Curriculum.

<https://curriculum.gov.bc.ca/curriculum/career-education/all/career-life-education>

Collins, A., Brown, J. S., & Newman, S. E. (1989). Cognitive apprenticeship: Teaching the crafts of reading, writing, and mathematics. In L. B. Resnick (Ed.), *Knowing, learning, and instruction* (pp. 453–494). Lawrence Erlbaum.

Jenkins, H., Purushotma, R., Weigel, M., Clinton, K., & Robison, A. J. (2009). *Confronting the challenges of participatory culture: Media education for the 21st century* (1st ed.). The MIT Press.

<https://doi.org/10.7551/mitpress/8435.001.0001>

Shafer, D. (2025). A critical thinking thematic framework and observation tool. *Thinking Skills and Creativity*, 56, Article 101787.